

Australia

Australian Competition & Consumer Commission

The Australian Competition and Consumer Commission (‘ACCC’) is an independent statutory authority established in 1995. Its role is to administer and enforce the *Competition and Consumer Act 2010* and other legislation, promoting competition, fair trading, and regulating national infrastructure for the benefit of all Australians.

The ACCC focusses on taking action that promotes the proper functioning of Australian markets, protects competition, improves consumer welfare and stops conduct that is anti-competitive or harmful to consumers.

The proliferation of data across the economy brings challenges but also opportunities. Key among those is the potential to detect competition and consumer issues sooner than before. As the ACCC expands its monitoring capabilities, and it refines methods to detect structural and behavioral anomalies, it increases the likelihood of detecting new issues as they emerge. The ACCC considers proactive detection to be a fundamental component of its growing strategic capability.

The Strategic Data Analysis Unit (SDAU) was established in 2017 to provide analytical advice and support to help the ACCC’s market inquiry work. In 2021, SDAU was combined with several other teams to form the Data & Intelligence Branch. Broadly speaking, the Data & Intelligence Branch helps other teams in the agency obtain and make sense of vast amounts of information for the purpose of making informed decisions. In addition to SDAU, the Data & Intelligence Branch includes: the Infocentre, the Intelligence Team and the Legal Technology Services Unit. The Data & Intelligence Branch works closely with the Information Management & Technology Services Branch to design technical solutions that help the agency address emerging challenges.

Below is a list of recent and current, and possible future, projects that the ACCC is working on.

I. Recent & current projects

A. Using Nuix Discover

- Ongoing use of predictive coding for document review in both investigation and litigation phases of matters;
- Concept clustering to group documents with common themes and identify connections between the cluster groups used to understand the themes across a document population in order to prioritize documents for review as well as to run quality checks on anomalies in the review;
- Trialing the creation of common terms across merger investigations and to use these as search term families as a starting point in prioritizing early case assessment and document review;
- Use of automated audio and video transcription tools to reduce time and cost and increase functionality in reviewing audio/video files;
- Use of compare tools to identify the differences in contract versions.

B. Other Artificial Intelligence and Machine Learning (and related) projects

- Using Natural Language Processing (NLP) techniques to automatically identify and measure groups of similar documents for investigations;
- Pattern-matching to identify entities (e.g. phone numbers, credit card numbers) across sets of documents;
- Entity recognition using NLP on incoming contacts to the agency, to better use aggregate data on the issues, products and companies of concern
- Topic modeling on documents and consumer complaints to analyze and visualize the common themes, allowing high-level insight into trends and possible areas for investigation;
- Web-scraping to collect information, including tools that mimic consumer behavior on websites;
- Online submissions portal with integrated computational support, including data validation;
- Cartel screening to detect possible bid-rigging in procurement;
- Automated searches of business records to identify persons of interests in cartel investigations.

II. Possible future projects

- Automated classification of incoming reports to speed up analysis and response, using natural language processing and a custom ML pipeline
- Internal predictive ML to estimate time and resourcing for projects and investigations, to support decisions about meeting organizational priorities
- Predictive coding for document review in both investigation and litigation phases of matters (within Nuix Discover);
- Concept clustering to group documents with common themes and identify connections between the cluster groups used to understand the themes across a document population in order to prioritize documents for review as well as to run quality checks on anomalies in the review (within Nuix Discover);
- Use of compare tools to identify the differences in contract versions etc. (within Nuix Discover);
- Combining Natural Language Processing (NLP) techniques with other techniques (vectorisation and clustering) to automatically identify and measure groups of similar documents for investigations;
- Pattern matching to identify phone numbers, credit card numbers across sets of documents;
- Entity recognition using NLP on incoming contacts to the agency, to better use aggregate data on the issue, products and companies of concern;
- Topic modelling on documents and consumer complaints to analyse and visualize the common themes, allowing high-level insight into trends and possible areas for investigation;
- Web-based data collection (web scraping) tools for advanced use to collect evidence, including tools to mimic consumer behavior on websites.

III. Near-future projects

- Automated classification of incoming complaints to speed up analysis and response, using natural language processing and a custom ML pipeline;
- Internal predictive ML to estimate time and resourcing for projects and investigations, to support decisions about meeting organisational priorities;
- Online submissions portal with integrated computational support, including data validation.